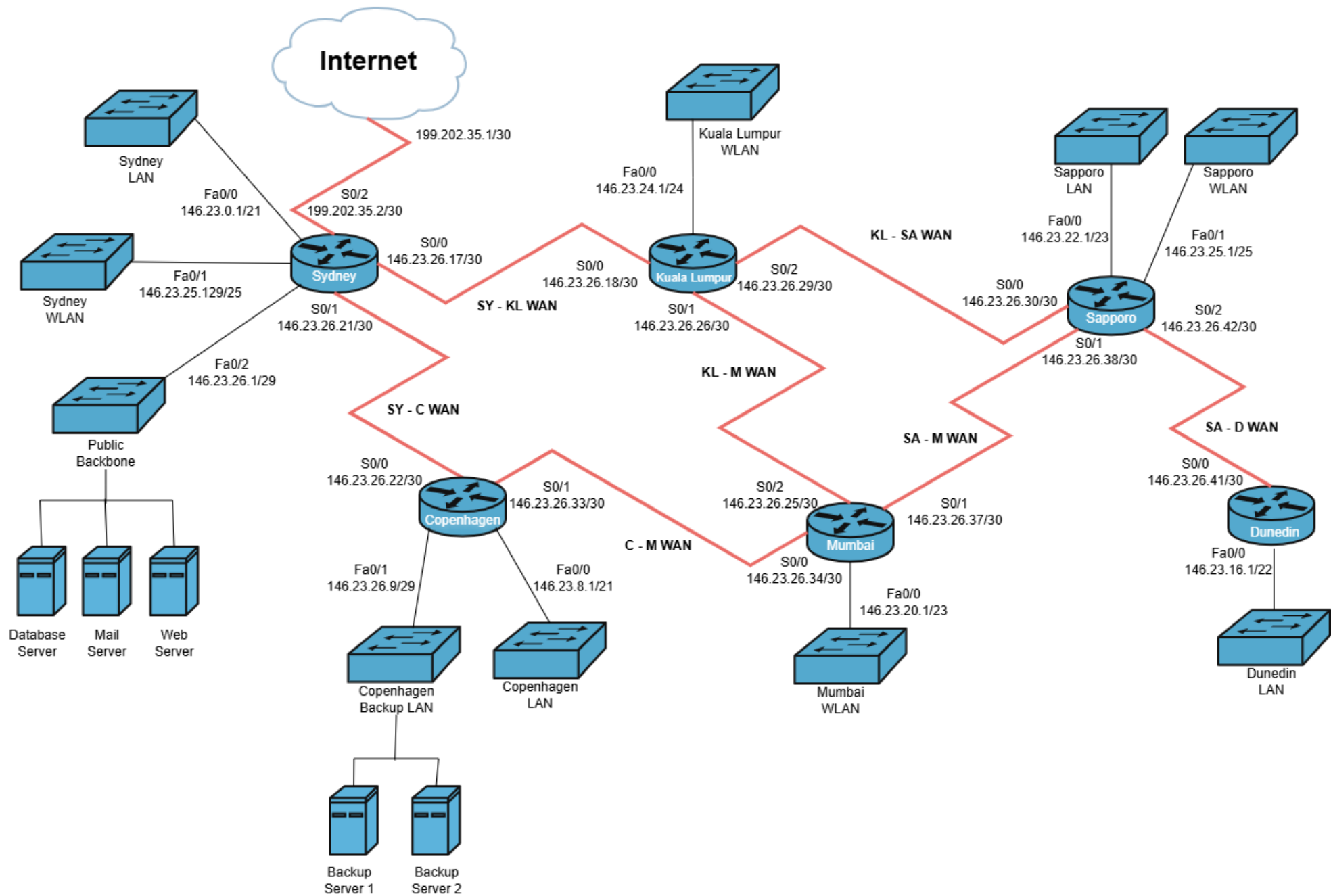




SUBNETS

Subnet Name	Subnet Address	Subnet Mask	First Usable Address	Last Usable Address	Broadcast Address	Static Address Range	DHCP Address Range
Sydney LAN	146.23.0.0	/21	146.23.0.1	146.23.7.254	146.23.7.255	146.23.0.1 - 146.23.0.10	146.23.0.11-146.23.7.254
Copenhagen LAN	146.23.8.0	/21	146.23.8.1	146.23.15.254	146.23.15.255	146.23.8.1 - 146.23.8.10	146.23.8.11-146.23.15.254
Dunedin LAN	146.23.16.0	/22	146.23.16.1	146.23.19.254	146.23.19.255	146.23.16.1 - 146.23.16.10	146.23.16.11-146.23.19.254
Mumbai WLAN	146.23.20.0	/23	146.23.20.1	146.23.21.254	146.23.21.255	146.23.20.1 - 146.23.20.10	146.23.20.11-146.23.21.254
Sapporo LAN	146.23.22.0	/23	146.23.22.1	146.23.23.254	146.23.23.255	146.23.22.1 - 146.23.22.10	146.23.22.11- 146.23.23.254
Kuala Lumpur WLAN	146.23.24.0	/24	146.23.24.1	146.23.24.254	146.23.24.255	146.23.24.1 - 146.23.24.10	146.23.24.11- 146.23.24.254
Sapporo WLAN	146.23.25.0	/25	146.23.25.1	146.23.25.126	146.23.25.127	146.23.25.1 - 146.23.25.10	146.23.25.11 - 146.23.25.126
Sydney WLAN	146.23.25.128	/25	146.23.25.129	146.23.25.254	146.23.25.255	146.23.25.129 - 146.23.25.138	146.23.25.139 - 146.23.25.254
Public Backbone	146.23.26.0	/29	146.23.26.1	146.23.26.6	146.23.26.7	146.23.26.1 - 146.23.26.6	N/A
Copenhagen Backup	146.23.26.8	/29	146.23.26.9	146.23.26.14	146.23.26.15	146.23.26.9 - 146.23.26.14	N/A
Sydney - Kuala Lumpur WAN	146.2.26.16	/30	146.23.26.17	146.23.26.18	146.23.26.19	146.23.26.17 - 146.23.26.18	N/A
Sydney - Copenhagen WAN	146.23.26.20	/30	146.23.26.21	146.23.26.22	146.23.26.23	146.23.26.21 - 146.23.26.22	N/A
Kuala Lumpur - Mumbai WAN	146.23.26.24	/30	146.23.26.25	146.23.26.26	146.23.26.27	146.23.26.25 - 146.23.26.26	N/A
Kuala Lumpur - Sapporo WAN	146.23.26.28	/30	146.23.26.29	146.23.26.30	146.23.26.31	146.23.26.29 - 146.23.26.30	N/A
Copenhagen - Mumbai WAN	146.23.26.32	/30	146.23.26.33	146.23.26.34	146.23.26.35	146.23.26.33 - 146.23.26.34	N/A
Mumbai - Sapporo WAN	146.23.26.36	/30	146.23.26.37	146.23.26.38	146.23.26.39	146.23.26.37 - 146.23.26.38	N/A
Sapporo - Dunedin WAN	146.23.26.40	/30	146.23.26.41	146.23.26.42	146.23.26.43	146.23.26.41 - 146.23.26.42	N/A

ROUTER INTERFACES

Loaction	Interface	IP address	Subnet Mask
Sydney	Fa0/0 (LAN)	146.23.0.1	/21
	Fa0/1 (WLAN)	146.23.25.129	/25
	Fa0/2 (Public Backbone)	146.23.26.1	/29
	S0/0 (WAN to Kuala Lumpur)	146.23.26.17	/30
	S0/1 (WAN to Coppenhagen)	146.23.26.21	/30
	S0/2 (Internet)	199.202.35.2	/30
Copenhagen	Fa0/0 (LAN)	146.23.8.1	/21
	Fa0/1 (Backup LAN)	146.23.26.9	/29
	S0/0 (WAN to Sydney)	146.23.26.22	/30
	S0/1 (WAN to Mumbai)	146.23.26.33	/30
Dunedin	Fa0/0 (LAN)	146.23.16.1	/22
	S0/0 (WAN to Sapporo)	146.23.26.41	/30
Mumbai	Fa0/0 (WLAN)	146.23.20.1	/23
	S0/0 (WAN to Coppenhagen)	146.23.26.34	/30
	S0/1 (WAN to Sapporo)	146.23.26.37	/30
	S0/2 (WAN to Kuala Lumpur)	146.23.26.25	/30
Sapporo	Fa0/0 (LAN)	146.23.22.1	/23
	Fa0/1 (WLAN)	146.23.25.1	/25
	S0/0 (WAN to Kuala Lumpur)	146.23.26.30	/30
	S0/1 (WAN to Mumbai)	146.23.26.38	/30
	S0/2 (WAN to Dunedin)	146.23.26.42	/30
Kuala Lumpur	Fa0/0 (WLAN)	146.23.24.1	/24
	S0/0 (WAN to Sydney)	146.23.26.18	/30
	S0/1 (WAN to Mumbai)	146.23.26.26	/30
	S0/2 (WAN to Sapporo)	146.23.26.29	/30

SERVERS

Location	Server Name	IP address	Subnet Mask
Sydney	Mail Server	146.23.26.2	/29
	Web Server	146.23.26.3	/29
	Database Server	146.23.26.4	/29
Copenhagen	Backup Server 1	146.23.26.10	/29
	Backup Server 2	146.23.26.11	/29

WEIGHTED DECISION MATRIX

Weight	Scores
0 = not relevant	0 = not met
1 = not very important	2 = partially
3 = must be met	4 = meets
5 = critical	6 = exceeds

ROUTERS

Routers						
Requirement	Cost	Performance/Throughput	Security	Scalability	Support/Warranty	Total Weighted Score
Description	Value within budget	Traffic Capacity	Security Features	Future Expansion	Reliable Vendor/Support	
Weight	1	5	5	3	3	
Score - Cisco Catalyst 8200	2	4	4	6	4	76
Score - Juniper SRX300	4	4	4	2	4	62
Score - Cisco ISR 1100	2	2	4	2	4	50
Score - Fortinet FortiGate 60F	6	6	6	4	4	90

Cost - Weight 1

The reasoning for this being a low weighted element is due to hardware decisions being dependent more so on value and suitability with justifiable leniency within the budget, prioritising exceptional security and performance over lower-cost options. After comparing prices, the most affordable option identified the Fortinet FortiGate 60F at approximately \$1,506 AUD and the most expensive Cisco ISR 1100 sitting at approximately \$2,148 while delivering the weakest overall score.

Performance/Throughput - Weight 5

Performance has the highest weight attached as it must be able to withstand the high traffic demands at this centralised site, therefore strong throughput is essential to avoid bottlenecks and performance drops. The Fortinet FortiGate 60F scores highest in this category by providing 10Gbps firewall throughput and 700Mbps IPsec throughput. These performance specs significantly outweigh its competitors; Juniper SRX300 (5 Gbps / 800 Mbps IPsec), Cisco Catalyst 8200L (3.8 Gbps / 500 Mbps IPsec), Cisco ISR 1100 (1.6 Gbps / 340 Mbps).

Security - Weight 5

Security is heavily weighted due to the essential protection of the network's infrastructure and privacy of sensitive data. The Fortinet FortiGate 60F receives the highest score exceeding competitors due to its fully integrated NGFW and UTM built directly within FortiOS without additional licensing requirements. Juniper SRX 300, Cisco Catalyst 8200 and Cisco ISR 1100 require additional licensing to achieve more advanced security features, while their base models do support some access to NGFW features.

Scalability – Weight 3

The scalability was weighted at three to balance existing networking needs with some consideration of future expansion. While expansion is important, business and networking needs may change over time. Therefore, the current needs of the company take priority, with scalability as a secondary factor. All devices did provide considerable scalability capabilities, with the FortiGate 60F allowing multi-site management and FortiOS SD-WAN features. However, the Cisco 8200 also offers SD-WAN with NIM expansion slots making it a considerable contestant, taking the highest score. The Juniper SRX300 and Cisco 1100 received lower scores due to requiring additional licensing but has SD-WAN capabilities.

Warranty – Weight 3

Warranty was given a lower weighting because although it is important to foster business with credible vendors and support systems, the industry standards applied across all devices with a 1-year warranty and basic technical support. There is minimal difference between warranty across all hardware making this section less impactful on final hardware decisions.

SWITCHES

Switches						
Requirement	Cost	Throughput	Security	Scalability	Port Density	Total Weighted Score
Description	Value within budget	Switching Capacity	Security Features	Future Expansion	Number of Ports	
Weight	1	5	5	3	5	
Score - Cisco Catalyst 9200-A	2	6	6	6	4	100
Score - Aruba CX 6200F 24G	2	6	6	6	2	90
Score - Juniper EX2300	4	6	6	4	4	96
Score - HPE Aruba 2930F	4	4	6	4	2	78

Cost - Weight 1

The reasoning for this being a low weighted element is due to hardware decisions being dependent more so on value and suitability with justifiable leniency within the budget, prioritising exceptional security, performance and port density over lower-cost options. The most affordable options are HPE Aruba 2930F (approx. \$1,800–2,200), and following behind is the Juniper EX2300 (approx. \$2,631). The Cisco Catalyst 9200 comes to a much higher price point at \$3,080, however the most expensive is the Aruba CX 6200F taking top place at approximately \$3,905.

Performance/Throughput - Weight 5

Performance has the highest weight attached as it must be able to withstand the high traffic demands at this centralised site, therefore strong throughput is essential to avoid bottlenecks and performance drops. Ranking highest in throughput are Cisco Catalyst 9200, Aruba CX 6200F, Juniper EX2300 – all respectively boasting 128 Gbps of switching capacity and 95.2 Mpps forwarding rate. In comparison the Aruba 2930F provides 56 Gbps switching capacity and 41.7 Mpps forwarding rate, making it less suitable for supporting enterprise-traffic.

Security - Weight 5

Security is heavily weighted due to the essential protection of the network's infrastructure and privacy of sensitive data. All devices ranked very highly in enterprise security with high end authentication, ACLs and network access controls. The Aruba CX 6200F again did have a slight edge with a more advanced and comprehensive security feature set, however all devices provided stronger than average security features.

Scalability - Weight 3

Scalability is weighted moderately to balance current network requirements with future expansion. The Cisco Catalyst 9200 and Aruba CX 6200F scored the strongest both supporting stacking of up to 8 switches for flexible expansion, using StackWise-160 and VSF stacking respectively. The Juniper EX2300 supports Virtual Chassis with up to 4 switches, and the Aruba 2930F also supports stacking however it is more limited in its VSF capabilities.

Port Density – Weight 5

Port density was highly weighted as the network must support many connected devices such as various hosts and WLANs. All switches compared are 24-port configurations, however differences in PoE budget are the key determining factors. Cisco Catalyst 9200 and Juniper EX2300 provide higher PoE budgets with 740W which is enough to power all 24 ports simultaneously, whereas the Aruba 2930F and Aruba CX 6200F (370W models) can sufficiently power 12 devices, only partially meeting requirements.

WAPs

WAPs						
Requirement	Cost	Throughput	Security	Scalability	Warranty	Total Weighted Score
Description	Value within budget	Wifi Traffic Capacity	Security Features	Management Flexibility	Reliable Vendor/Support	
Weight	1	5	5	3	5	
Score - Cisco Catalyst 9130AX	0	6	6	6	6	108
Score - HPE Aruba AP-515	4	6	6	6	6	112
Score - Ubiquiti UniFi U6 Pro	6	4	4	4	4	78
Score - Ruckus R550	4	2	4	2	6	60

Cost - Weight 1

The reasoning for this being a low weighted element is due to hardware decisions being dependent more so on value and suitability with justifiable leniency within the budget, prioritising exceptional security, performance and warranty over lower-cost options. The Ubiquiti UniFi U6 Pro is the most affordable option at approximately \$370 AUD per unit, Ruckus R550 \$575 per unit, and HPE Aruba AP-515

\$1041 per unit, and finally the Cisco Catalyst 9130AX approx. \$2,627 AUD per unit. Purchasing four units alone would exceed the budget of \$10,000 making it the most unsustainable option with a score of 0.

Performance/Throughput - Weight 5

Performance has the highest weight attached as it must be able to withstand the high traffic demands at this centralised site, simultaneously supporting various wireless users with consistent performance. All devices use Wi-Fi 6 -the current enterprise standard – but the devices with the highest output performance include the HPE Aruba AP-515 (5.40 Gbps) and similarly the Cisco Catalyst 9130AX (5.38 Gbps). The Ubiquiti U6 Pro (5.3 Gbps) is comparable in terms of speed but lacks scanning radio or RF management of the Cisco and Aruba platforms, lowering its score slightly. The Ruckus shows the least suitability with a score of 2 with 1.77 Gbps, ultimately not able to withstand user demands in a high-traffic site.

Security - Weight 5

Security is heavily weighted due to the essential protection of the network's infrastructure and privacy of sensitive data. The Cisco Catalyst 9130AX and HPE Aruba AP-515 both scored highly with strong security features such as WPA3, 802.1X/RADIUS user authentication, VLAN assignment, segmentation/tagging etc. The Ubiquiti UniFi U6 Pro and the Ruckus R550 supports standard security features but does not exceed to the same degree of their competitors.

Scalability - Weight 3

Scalability is weighted moderately to balance current network requirements with future expansion while remaining easily manageable. The Cisco Catalyst 9130AX and HPE Aruba AP-515 both offer strong, flexible scalability with AI features allowing easy centralised enterprise-wide deployment. Ubiquiti U6 Pro also offered satisfactory scalability but lacked the same level of optimisation, and Ruckus R550 is better suited for smaller-scaled deployments with lacking cloud analytics and centralised RF management giving it the lowest score.

Warranty – Weight 5

Warranty is rated high priority due to the amount of per units of WAPs required to purchase, ensuring coverage of a large sum of devices and support for long term usage. Score 6 was given to all devices that had Limited Lifetime Hardware Warranty, which exceeds the industry standard of 1 year warranty as seen in the Ubiquiti U6 Pro – hence it's score of 4. Due to the overall similarity in warranty results this does not have significant influence on the hardware decision making process.

BUDGET

Device	Qty	Cost (per unit)	Total
Fortinet FortiGate 60F	1	\$1,506	\$1,506
Cisco C9200-24P-A	2	\$3,080	\$6,158
HPE Aruba AP-515	4	\$1,041	\$4,164
Total			\$11,830

To meet the requirements based on Sydney's topology we will require 2x switches dedicated to its backbone and 1020 workstations LANs, 4x WAPs to support 80 WLAN users and 1 router that centrally manages all connections. Referring to each decision matrix it is possible to identify the most suitable candidates based on the total weighted score and additional considerations mentioned in the justifications. In this instance, out of the devices compared the most suitable devices to deploy would be the Fortinet FortiGate 60F router, 2x Cisco C9200-24P-A switches, and 4x HPE Aruba AP-515 WAPs. Budget wise this does exceed the allocated amount of \$10,000 by \$1,830; however, this can be a justified surplus as the devices meet all required enterprise levels of performance, security, scalability and port density priorities with minimal budget overdraw. While it is worth noting the Juniper EX2300 was a cheaper alternative comparable to Cisco Catalyst 9200, Cisco's PoE and stacking capabilities were a defining factor between the two. As for Fortinet FortiGate 60F it proved to be the cheapest and highest performing, and the HPE Aruba AP-515 had the strongest overall score with consistent performance within a suitably medium-ranged price tier.